

Learning Resource On Speech and Video Processing

Module-1: Speech Processing Concepts

Chapter-2: Digital Representation of Speech Signals

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Chapter Outcomes

- After the completion of this chapter, the students will be able to:
 - Explain the process of converting analog speech signals into digital form using sampling techniques.
 - Describe the role of quantization and encoding in the digital representation of speech signals.
 - Illustrate the complete process of digital speech representation using sampling, quantization, and encoding.

- A *signal* is defined as any physical quantity that varies with time, space, or any other independent variable or variables.
- Speech is naturally an analog signal, where sound waves vary continuously with time.
- Modern communication and computing systems, however, process speech in digital form.
- Digital representation makes speech easier to store, transmit, analyze, and process using computers and digital devices.
- Digital representation converts continuous speech into a sequence of binary numbers that computers can efficiently process and analyze.

- A time-series signal is a function of time.

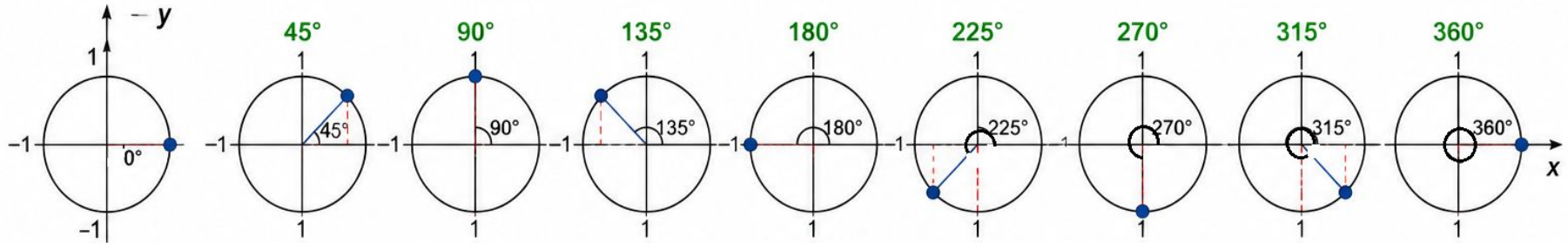
$$\text{e.g. } s_1(t) = 10t, s_2(t) = 5t^2$$

- The simplest periodic signal is a sine wave and can be represented as

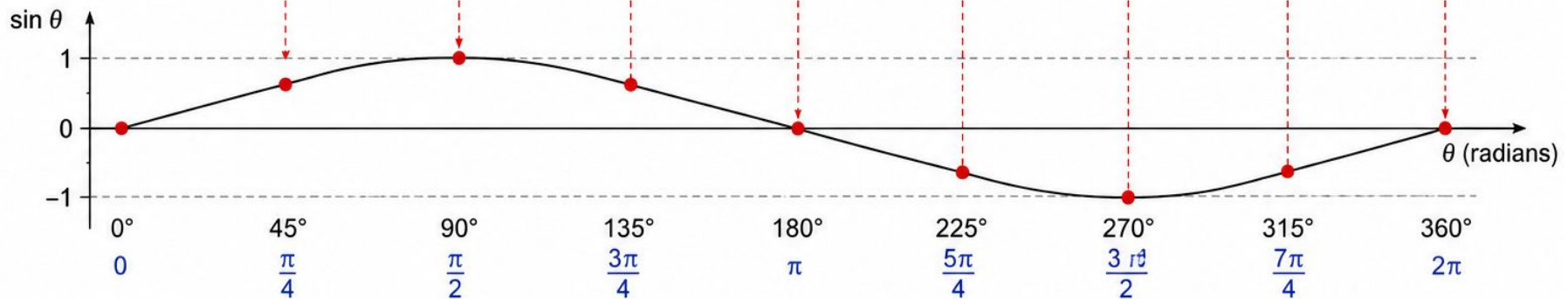
$$x(t) = A \sin(2\pi ft + \phi)$$

- where
 - $x(t)$ = instantaneous value (signal amplitude) at time t
 - A = amplitude (maximum value of the wave),
 - f = frequency (Hz),
 - t = time (seconds),
 - ϕ = phase angle (radians)

1. Motion on the Unit Circle



2. Vertical Projection ($y = \sin \theta$)

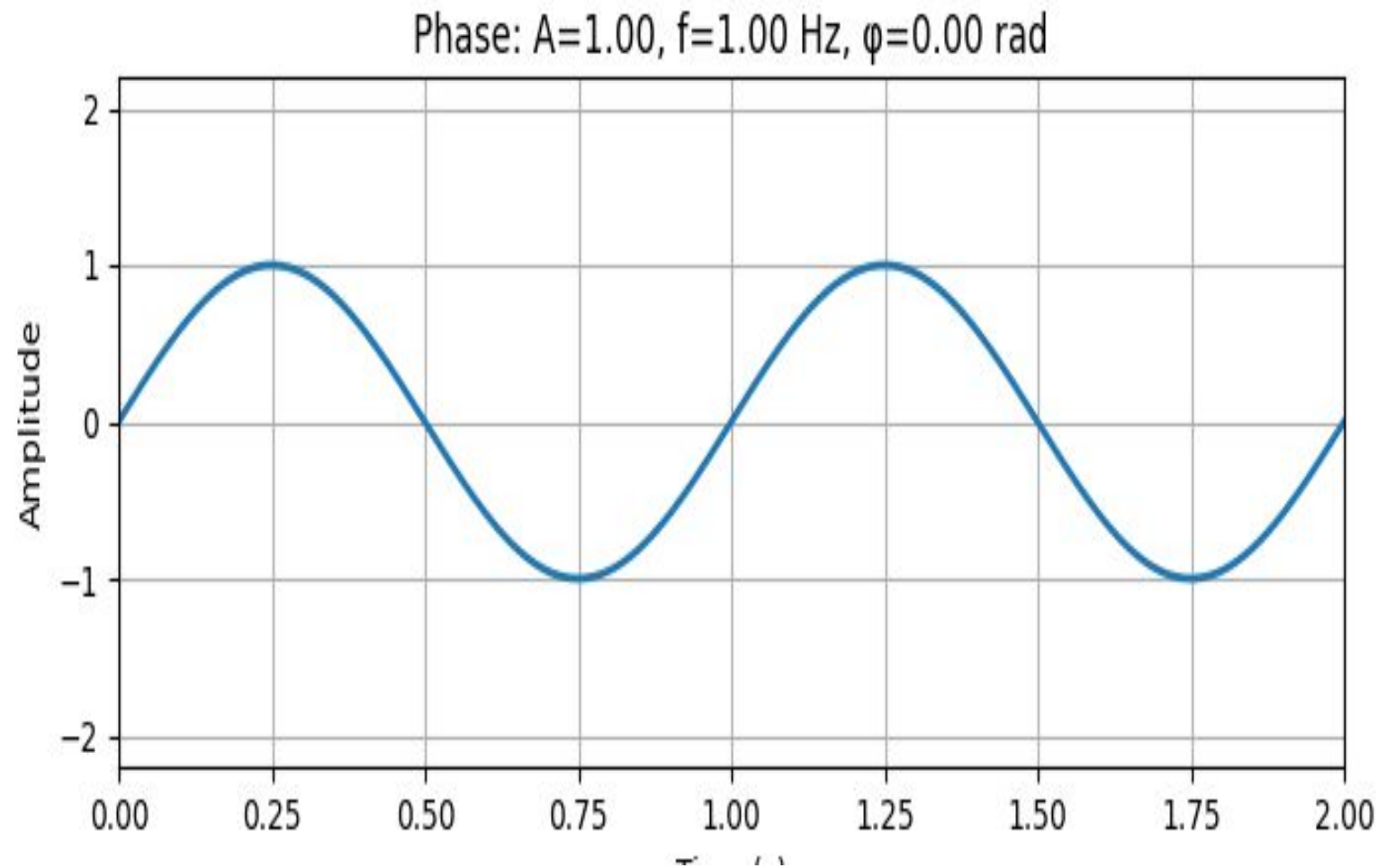


3. Mapping Table

θ (degrees)	0°	45°	90°	135°	180°	225°	270°	315°	360°
θ (radians)	0	$\pi/4$	$\pi/2$	$3\pi/4$	π	$5\pi/4$	$3\pi/2$	$7\pi/4$	2π
$\sin \theta$ (y)	0	$\sqrt{2}/2$	1	$\sqrt{2}/2$	0	$-\sqrt{2}/2$	-1	$-\sqrt{2}/2$	0

As a point P moves counterclockwise on the unit circle with angle θ , its vertical projection on the y-axis is $\sin \theta$.
Plotting this projection versus θ gives the sine wave.

contd...



- Any natural (e.g. speech) signal can be approximated to arbitrary accuracy as **a sum of sinusoidal components having different frequencies, amplitudes, and phases.**

$$x(t) = \sum_{k=1}^{\infty} A_k \sin(2\pi f_k t + \phi_k).$$

where

A_k = amplitude,

f_k = frequency,

ϕ_k = phase.

Detailed discussion in *Fourier Transform*.

Continuous-Time VS Discrete-Time Signals

- A **continuous-time signal** is a signal that is defined for **every instant of time** within a given interval. When defining continuous sinusoidal signal $x(t)$, t is a continuous variable

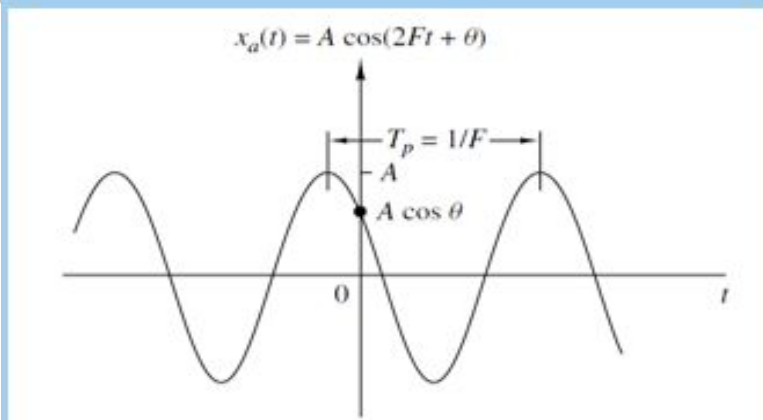
$$x(t) = A \sin(\omega_c t + \phi), \quad -\infty < t < \infty$$

where, ω_c (*Angular Frequency*) = $2\pi f$ (*rad/s*)

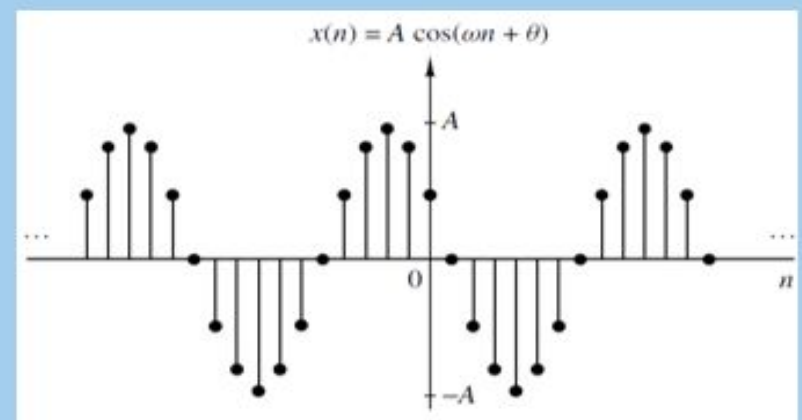
- A **discrete-time signal** is a signal whose value is defined only at **discrete instants of time**. It is represented as $x[n]$, where n is an integer index.
- A discrete-time sinusoidal signal may be expressed as

$$x[n] = A \sin(\omega n + \phi), \quad n = 0, \pm 1, \pm 2, \dots$$

where, ω = **digital angular frequency**(*rad/sample*)



Example of an analog sinusoidal signal.



Example of a discrete-time sinusoidal signal ($\omega = \pi/6$, and $\theta = \pi/3$).

Continuous-Time Signal

Represented as **(x(t))**

Defined for **every** instant of time

Time is a **continuous** variable

Example: Human speech before recording, Voltage across an electrical circuit

Discrete-Time Signal

Represented as **(x[n])**

Defined only at **discrete** sampling instants

Time is represented by an **integer** sample index

Example: Recorded speech sampled at 16 kHz

Continuous-Valued vs Discrete-Valued Signals

- If a signal takes on all possible values on a finite or an infinite range, it is said to be a **continuous-valued** signal.
- Alternatively, if the signal takes on values from a **finite set of possible values**, it is said to be a **discrete-valued** signal.
- A **discrete-time signal having a set of discrete values** is called a *digital signal*.

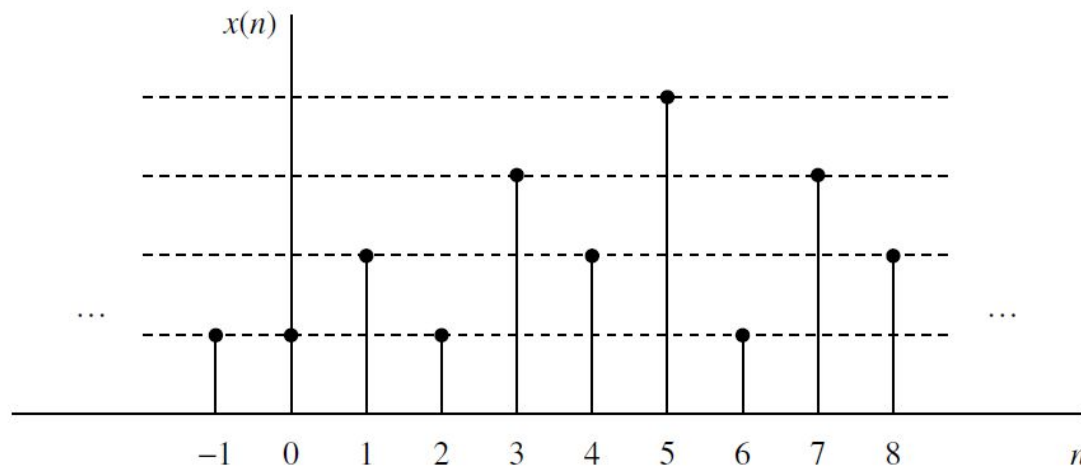


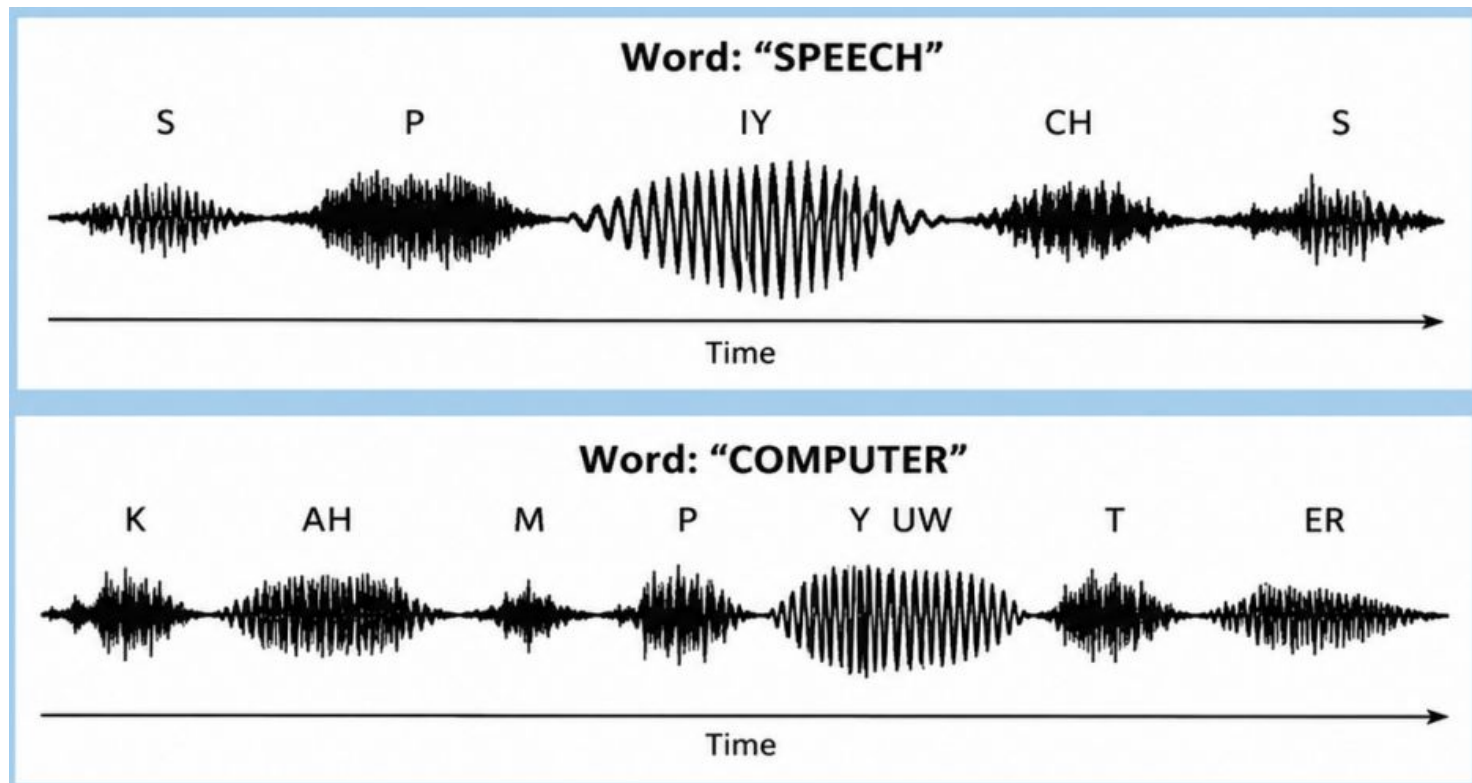
Figure 2.5 Digital signal with four different amplitude values.

Digitization of Speech

- **Speech** is an acoustic wave. A source vibrates (e.g. vocal cords, speaker). The vibration causes nearby particles to move back and forth, creating regions of compression and expansion. The disturbance in the particle travels as acoustic wave.
- It is a **non-stationary signal**, meaning its statistical properties change over time.
- It is **quasi-periodic** in nature, with voiced sounds showing periodicity due to vocal cord vibrations, while unvoiced sounds resemble noise.
- **Characteristics of Acoustic Waves**
 - **Frequency(f)**: determines the **pitch** of the sound (Hz)
 - **Amplitude(a)**: determines the **loudness** of the sound.
 - **Wavelength (λ)**: distance between successive compressions.

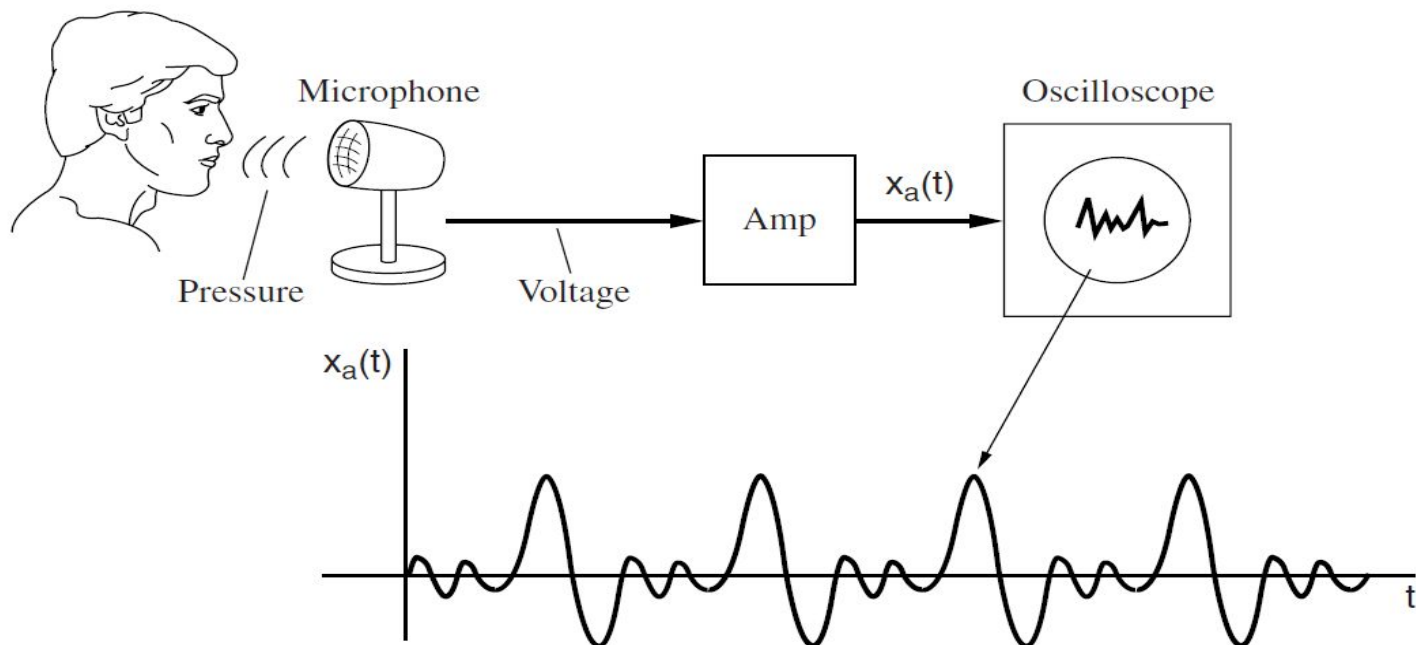
- A **speech signal** is a continuous acoustic waveform produced when a person speaks.
- It carries linguistic information such as
 - **phonemes**: Smallest unit of speech (speech sounds),
 - **syllables**: single unit of pronunciation in a word
 - **words**: unit of language made of one or multiple syllable where each syllable consists of one or more phonemes.
 - Ex: **Word: “Speech”**
 - **Phonemes**: /s/ – /p/ – /i:/ – /tʃ/
 - **Syllable**: one syllable
 - **Word: “Computer”**
 - **Syllables**: com-pu-ter (3 syllables)
 - **Phonemes**: /k/ /ə/ /m/ /p/ /j/ /u:/ /t/ /ər/

- In speech signal, variability arises from speaker differences (age, gender, accent), speaking rate, and background noise. This variability makes speech representation challenging but also rich in information.



Digitization

- The pressure wave from an acoustic can be transduced to an electrical signal with a microphone.
- The microphone voltage can be amplified and viewed on an oscilloscope.
- The resulting analog waveform is denoted by $x_a(t)$, a *continuous-time signal* (an *analog waveform*).



- For digital speech processing input, acoustic signal is converted to electrical signal (microphone).
- For digital speech processing output, electrical signal is to be converted to acoustic signal (speaker).

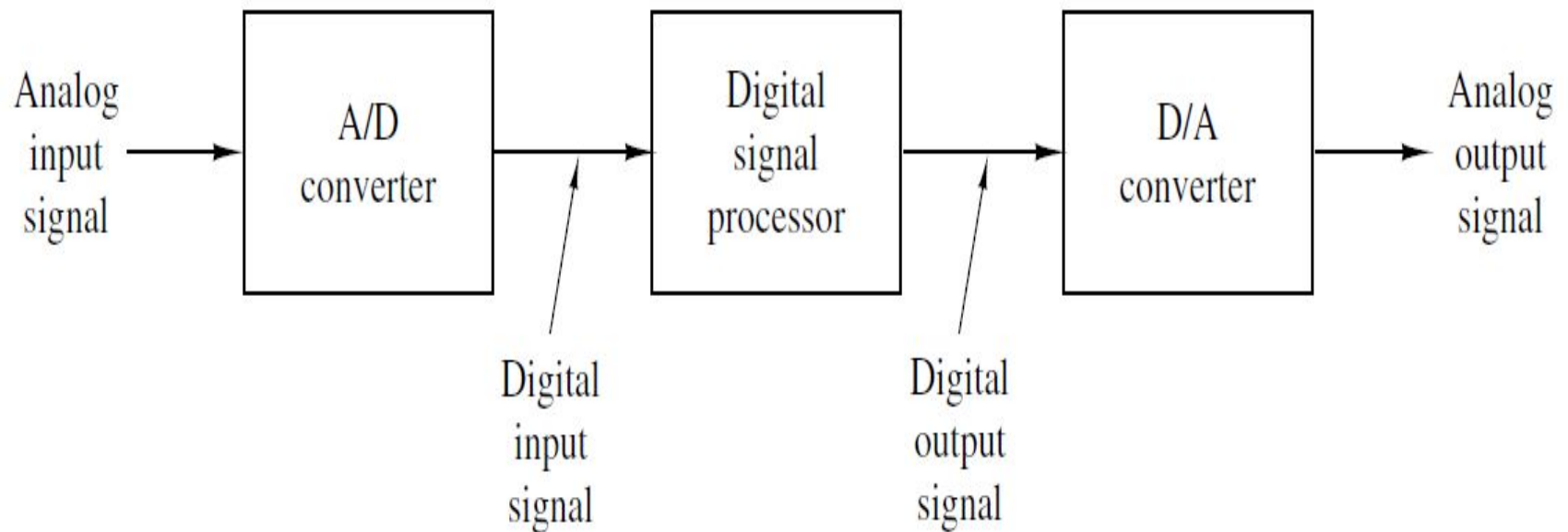
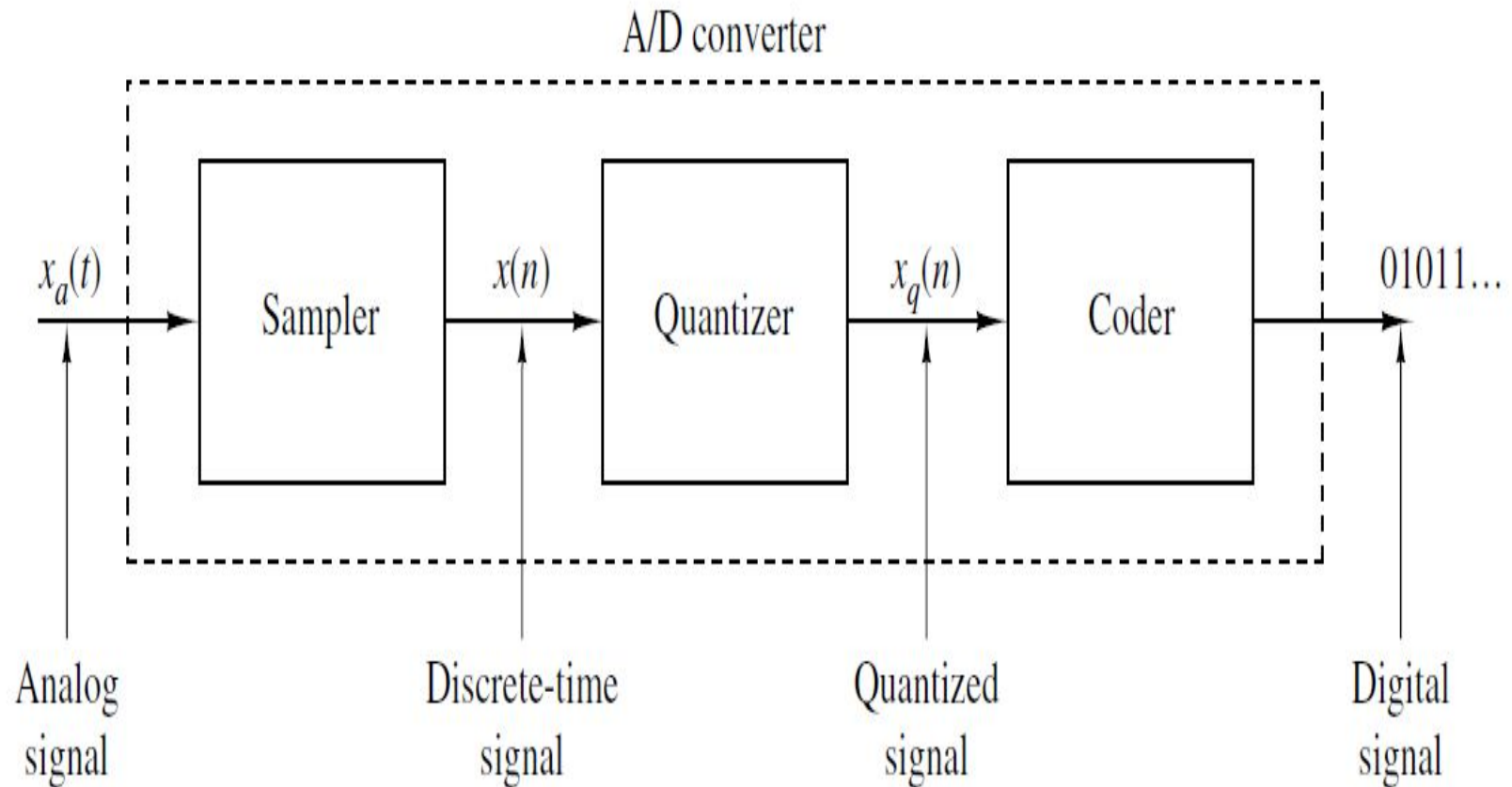


Figure 1.3 Block diagram of a digital signal processing system.

- The process to convert analog signals into digital form is called *analog-to-digital (A/D) conversion*, and the corresponding devices are called *A/D converters (ADCs)*.
- **Sampling:** Taking “samples” of the continuous-time signal at discrete time instants.
- **Quantization:** Conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal.
- **Coding:** In the coding process, each discrete value $x_q(n)$ is represented by a b -bit binary sequence



- **Sampling** is the process of measuring the amplitude of an analog speech signal at regular time intervals to obtain a discrete-time (digital) representation.
- Sampling allows the computers to perform various operations on speech.
- $x_a(t)$ is *sampled* at uniformly spaced time instants with T being the **sampling interval**.
- $1/T$ is called as **sampling frequency** (F_s).
- **Output** of the sampler (a *continuous-to-discrete (C/D) converter*) is a series of numbers $x_a(nT)$ whose representation is simplified as

$$x[n] = x_a(nT)$$

- The series $x[n]$ is referred to as a **discrete-time signal** or

Sampling Terminologies

- **Sampling Frequency (F_s):** Number of samples taken every second.
 - Unit: Hertz (Hz)
 - Examples: 8000 Hz; 16000 Hz; 44100 Hz
- **Sampling Interval (T_s):** Time between two consecutive samples.

$$T_s = \frac{1}{F_s}$$

- **Example:** If F_s is 8000 Hz then sampling interval T_s will be

$$T_s = \frac{1}{8000} = 0.000125 \text{ s} = 125\mu\text{s}$$

- **Bandwidth:** Speech does not contain infinitely high frequencies.
- The bandwidth determines the minimum sampling rate.
- Common speech bandwidths are: Telephone (300–3400 Hz); Wideband (50–7000 Hz); High quality audio (20–20000 Hz).

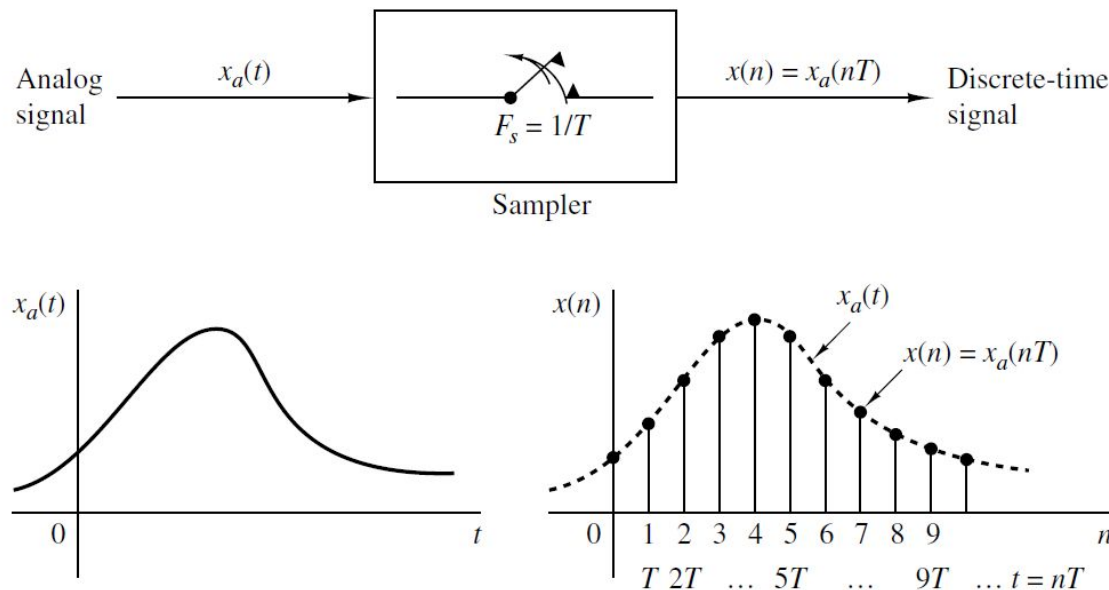


Figure 4.3 Periodic sampling of an analog signal.

Nyquist–Shannon Sampling Theorem

- Nyquist-Shannon Sampling theorem is one of the most important principles in digital signal processing.
- This theorem states that “To perfectly reconstruct a continuous-time signal from its samples without distortion, the sampling rate must be at least *twice the highest frequency* present in the signal.”
- **Sampling rate $(f_s) \geq 2 \times$ maximum signal frequency (f_{max})**
- This minimum rate is called the ***Nyquist rate* $(2f_{max})$** , and half of the sampling frequency is the ***Nyquist frequency* $(f_s/2)$** .

- Human hearing extends up to ~ 20 kHz. CDs use **44.1 kHz sampling rate**, slightly above 2×20 kHz, to ensure faithful reproduction.
- Telephony systems often use **8 kHz sampling rate** (Nyquist frequency = 4 kHz), which is enough for intelligible voice transmission.
- If sampling frequency is too low, then high-frequency details are missed and **aliasing** occurs.
- **Aliasing:** During sampling, the spectrum of the original signal is repeated at multiples of the sampling frequency. If the sampling frequency is too low, these repeated spectra overlap. This overlap causes different frequencies to become indistinguishable, leading to distortion known as aliasing.
- Due to aliasing, high-frequency components are incorrectly interpreted as lower frequencies, causing distortion and making the original signal impossible to recover accurately.

Example: Suppose the speech signal contains frequencies up to 5000 Hz.

- If the sampling happens at 12000 Hz and the Nyquist rate is $2 \times 5000 = 10000$ Hz then there won't be aliasing as the sampling frequency is way above the Nyquist rate.
- However, the sampling frequency is 6000Hz then aliasing will happen as the sampling frequency is below the Nyquist rate which will create problems in capturing the highest frequency components.
- Aliasing can be prevented by:
 - Sampling **at or above the Nyquist rate**.
 - Using an **anti-aliasing filter (a low-pass filter)** before sampling to remove frequencies above the Nyquist frequency.

Quantization

- Quantization is the process **to convert continuous amplitudes** into **discrete levels** so signals can be encoded in binary.
- The purpose of quantization is to make a continuous-amplitude signal representable in the digital domain by restricting it to a finite set of discrete values.
- Without quantization, sampled signals would have infinitely many possible amplitudes, making them impossible to store or process digitally.
- It reduces the amount of information needed to represent a signal, enabling efficient storage and transmission.
- It is done after sampling, during Analog-to-Digital Conversion (ADC), and introduces a small error called **quantization noise/error**.

Quantization Step Size (QSS)

- **Quantization Step Size (Δ)** : Difference in amplitude between two successive quantization levels.
- Mathematically, quantization step size can be represented as below:

$$\Delta = \frac{V_{max} - V_{min}}{L}$$

where,

V_{max} = Maximum value of the signal

V_{min} = Minimum value of the signal

$L = 2^N$; where, N = No. of bits per sample

Bit rate and storage

- **Sampling** converts a continuous-time signal into discrete-time samples.
- **Quantization** converts each sample's amplitude into one of a limited number of allowed values.
- **Encoding** assigns binary codes to these quantized levels for digital representation.
- Based on the sampling and quantization, storage size will be affected.

$$\text{File Size (bits)} = f_s \times N \times T; \text{ bit rate (bps)} = f_s \times N$$

where, f_s = **sampling frequency** (samples per second); N = **bits per sample** (quantization resolution); T = **duration of recording** (seconds)

$$T = \frac{\text{File Size (bits)}}{f_s \times N}$$

Example

- A speech signal is uniformly quantized using **8 bits per sample** over a range of $\pm 2 \text{ V}$. Find the **quantization step size (Δ)**. If the sampling frequency 10 Hz, then what will be **the size of the file for a 10 sec recording?**

Solution: Number of levels $L = 2^N = 2^8 = 256$

Range = 4 V (from -2 V to +2 V)

$$\Delta = \frac{\text{Range}}{L} = \frac{4}{256} = 0.015625 \text{ V}$$

So, Step size = **0.0156 V**

Sampling frequency (F_s) = 10 Hz i.e., 10 samples per second

Each sample = 8 bits = 1 byte

Data per second = $10 \times 1 \text{ byte} = 10 \text{ bytes}$

- A speech signal is uniformly quantized using 16 bits per sample over a range of ± 5 V. Find the quantization step size (Δ). If the sampling frequency 12 Hz, then what will be the size of the file for a 10 sec recording?
- Ans: Step size = 0.000152 V
File size = 240 bytes

Quantization Error

- Quantization error is the **difference between the original sample value and its quantized (rounded) value.**

$$e[n] = x[n] - x_q[n]$$

where,

$x[n]$ = original sampled signal

$x_q[n]$ = quantized signal

$e[n]$ = quantization error (or quantization noise)

- **Maximum Quantization Error:** For a **uniform rounding quantizer**, the error always lies between $-\frac{\Delta}{2} \leq e[n] \leq \frac{\Delta}{2}$, where Δ is the **quantization step size**.
- **Mean squared Quantization Error:** If the quantization error is assumed to be uniformly distributed, the average quantization noise power is

$$\Delta^2/12$$

Suppose there is a 8-bit ADC with input range: -1 V to $+1\text{ V}$. Find the Quantization Step size, Maximum Quantization Error and Mean squared Quantization Error.

Solution:

No. of bits = 8

So, No. of level (L) = $2^N = 2^8 = 256$

Given Voltage range = 2 V

QSS (Δ) = $2 / 256 = 0.0078125\text{ V}$

Max. Quantization Error ($e_{\max}$) = $\Delta / 2 = 0.0078125 / 2$
 $= 0.00390625\text{ V}$

Mean squared Quantization Error (σ_e^2) = $\Delta^2 / 12$
 $= (0.0078125)^2 / 12$

Practice Problems

- A 10-bit ADC converts analog signals in the range -5 V to $+5\text{ V}$. Determine Quantization step size, maximum quantization error, and mean squared quantization error.
- Suppose the ADC is modified to operate with a 12-bit resolution instead of 10-bit resolution. Discuss how this change affects the ADC's operation.
- A speech acquisition system samples a signal at 16 kHz using a 10-bit ADC over the input range -1.5 V to $+1.5\text{ V}$. Calculate
 - Quantization step size
 - Maximum quantization error
 - Mean squared quantization error